

Key Knowledge and future improvements

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Future Improvements:

- I would like to optimize the pipeline to reduce processing time per frame, getting the system much closer to true real-time detection.
- The current bot only sends text messages. I would like to expand its functionality to automatically make calls to emergency authorities and include the GPS coordinates of the location.
- I would explore better hyperparameter configurations or expand the dataset to achieve a better recall.

Challenges:

- One of the biggest challenges during training was that Google Colab kept closing my environment, interrupting the training process. To solve this, I connected the project to Google Drive. This way, no progress was lost regardless of how many times the session closed.
- Building the Telegram alerting system was challenging because I could not find any tutorial or documentation that matched exactly what I was trying to build. I asked Claude for help, and it provided me with a direct link to the official Telegram Bot API documentation, which had everything I needed.